EE 5239 Nonlinear Optimization

Lecture 14b: The Alternating Direction Method of Multipliers (II)

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Outline

Announcement

- HW 6 is posted and due Monday Dec 16th
- Lecture on Dec 11th will be canceled
- JL's office hour will be moved to Monday Dec 9th 4-5pm

Last Time

- Introduced the ADMM Algorithm
- Consider the Equality-Constrained convex problem

$$\min_{\boldsymbol{x}\in X, \mathbf{z}\in Z} \quad f(\boldsymbol{x}) + g(\mathbf{z}), \quad \text{s.t. } \mathbf{A}\boldsymbol{x} + \mathbf{B}\mathbf{z} = \mathbf{c}$$

- Augmented Lagrangian

$$L_\rho(\boldsymbol{x}, \mathbf{z}; \mathbf{y}) = f(\boldsymbol{x}) + g(\mathbf{z}) + \langle \mathbf{y}, \mathbf{A}\boldsymbol{x} + \mathbf{B}\mathbf{z} - \mathbf{c} \rangle + \frac{\rho}{2}\|\mathbf{A}\boldsymbol{x} + \mathbf{B}\mathbf{z} - \mathbf{c}\|^2$$

- Method of multipliers

$$\begin{aligned} \left(\boldsymbol{x}^{r+1}, \mathbf{z}^{r+1}\right) &= \arg\min_{\boldsymbol{x},\mathbf{z}} L_\rho(\boldsymbol{x}, \mathbf{z}; \mathbf{y}^r) \\ \mathbf{y}^{r+1} &= \mathbf{y}^r + \rho\left(\mathbf{A}\boldsymbol{x}^{r+1} + \mathbf{B}\mathbf{z}^{r+1} - \mathbf{c}\right) \end{aligned}$$

This Time

- The ADMM algorithm
- Go over a few applications, and see how ADMM iteration works
- Go over in detail the video background/foreground separation problem
- And the MATLAB codes
- See how the algorithm works in practice

Stephen Boyd's book `https://web.stanford.edu/~boyd/papers/pdf/admm_distr_stats.pdf`

Outline

The ADMM Algorithm

- Consider the Equality-Constrained convex problem

$$\min_{\boldsymbol{x} \in X, \mathbf{z} \in Z} \quad f(\boldsymbol{x}) + g(\mathbf{z}), \quad \text{s.t. } \mathbf{A}\boldsymbol{x} + \mathbf{B}\mathbf{z} = \mathbf{c}$$

- Augmented Lagrangian

$$L_\rho(\boldsymbol{x}, \mathbf{z}; \mathbf{y}) = f(\boldsymbol{x}) + g(\mathbf{z}) + \langle \mathbf{y}, \mathbf{A}\boldsymbol{x} + \mathbf{B}\mathbf{z} - \mathbf{c} \rangle + \frac{\rho}{2} \|\mathbf{A}\boldsymbol{x} + \mathbf{B}\mathbf{z} - \mathbf{c}\|^2$$

- Method of multipliers

$$\begin{aligned} \left(\boldsymbol{x}^{r+1}, \mathbf{z}^{r+1}\right) &= \arg\min_{\boldsymbol{x}, \mathbf{z}} L_\rho(\boldsymbol{x}, \mathbf{z}; \mathbf{y}^r) \\ \mathbf{y}^{r+1} &= \mathbf{y}^r + \rho \left(\mathbf{A}\boldsymbol{x}^{r+1} + \mathbf{B}\mathbf{z}^{r+1} - \mathbf{c}\right) \end{aligned}$$

The ADMM Algorithm

The steps of the ADMM Algorithm is given below

$$
\begin{aligned}
\boldsymbol{x}^{r+1} &= \arg\min_{\boldsymbol{x}\in X} L_\rho(\boldsymbol{x}, \mathbf{z}^r; \mathbf{y}^r) \\
\mathbf{z}^{r+1} &= \arg\min_{\mathbf{z}\in Z} L_\rho(\boldsymbol{x}^{r+1}, \mathbf{z}; \mathbf{y}^r) \\
\mathbf{y}^{r+1} &= \mathbf{y}^r + \rho\left(\mathbf{A}\boldsymbol{x}^{r+1} + \mathbf{B}\mathbf{z}^{r+1} - \mathbf{c}\right)
\end{aligned}
$$

- Divide and conquer: Optimize $\boldsymbol{x}$ and $\mathbf{z}$ once (coordinate descent on L_ρ), then update the dual variable
- The primal problem is no longer solved exactly (where the efficiency comes from)

The ADMM Algorithm

- One can view it as updating three variables (2 primal, 1 dual) alternatingly
- We can switch the role of two primal variables $\boldsymbol{x}$ and $\mathbf{z}$
- There is no restriction on the stepsize ρ, at long as $\rho > 0$

Optimality Conditions (READ)

- Lagrangian

$$L(\boldsymbol{x}, \mathbf{z}; \mathbf{y}) = f(\boldsymbol{x}) + g(\mathbf{z}) + \langle \mathbf{y}, \mathbf{A}\boldsymbol{x} + \mathbf{B}\mathbf{z} - \mathbf{c} \rangle$$

- KKT condition (suppose X, Z are both the whole space, no other constraints on $\boldsymbol{x}$, $\mathbf{z}$)

$$\begin{aligned} & -\mathbf{A}^T\mathbf{y}^* \in \partial f(\boldsymbol{x}^*), \quad -\mathbf{B}^T\mathbf{y}^* \in \partial g(\mathbf{z}^*) \\ & \mathbf{A}\boldsymbol{x}^* + \mathbf{B}\mathbf{z}^* - \mathbf{c} = 0 \end{aligned}$$

- ADMM updates (optimality condition at each iteration)

$$\begin{aligned} -\mathbf{A}^T\mathbf{y}^{r+1} + \mathbf{A}^T\left(\mathbf{B}\mathbf{z}^{r+1} - \mathbf{B}\mathbf{z}^r\right) &\in \partial f(\boldsymbol{x}^{r+1}) \\ -\mathbf{B}^T\mathbf{y}^{r+1} &\in \partial g(\mathbf{z}^{r+1}) \end{aligned}$$

- Optimality is achieved if the following are satisfied:

$$\begin{aligned} \mathbf{A}^T\left(\mathbf{B}\mathbf{z}^{r+1} - \mathbf{B}\mathbf{z}^r\right) &= 0 \\ \mathbf{A}\boldsymbol{x}^{r+1} + \mathbf{B}\mathbf{z}^{r+1} - \mathbf{c} &= 0 \end{aligned}$$

Convergence

- The ADMM converges under very mild condition
- Let us define the residue as

$$\mathbf{r}^r = \mathbf{A}\boldsymbol{x}^r + \mathbf{B}\mathbf{z}^r - \mathbf{c}$$

- **Claim**: Suppose that the problem is convex and feasible, then the following is true

  1. **Residue convergence**: $\mathbf{r}^k \to 0$ as $k \to \infty$
  2. **Objective convergence**: $f(\boldsymbol{x}^k) + g(\mathbf{z}^k) \to p^*$ as $k \to \infty$
  3. **Dual variable convergence**: $\mathbf{y}^k \to \mathbf{y}^*$ as $k \to \infty$, where $\mathbf{y}^*$ is the optimal dual solution

- Note $\boldsymbol{x}^r$ and $\mathbf{z}^r$ need not converge to optimal solutions, although such results can be shown under additional assumption

Convergence Rate

- Let $\bar{\boldsymbol{x}}^T = \frac{1}{T}\sum_{r=1}^{T} \boldsymbol{x}^t$ (averaging over all past iterates)
- Let $\bar{\mathbf{z}}^T = \frac{1}{T}\sum_{r=1}^{T} \mathbf{z}^t$ (averaging over all past iterates)
- Convergence rate for the objective

$$f(\bar{\boldsymbol{x}}^T) + g(\bar{\mathbf{z}}^T) - [f(\bar{\boldsymbol{x}}^*) + g(\bar{\mathbf{z}}^*)] \leq \mathcal{O}\left(\frac{1}{T}\right)$$

- Convergence rate for optimality conditions

$$\|\mathbf{A}\boldsymbol{x}^T + \mathbf{B}\mathbf{z}^T - \mathbf{c}\|^2 + \|\mathbf{B}\mathbf{z}^T - \mathbf{B}\mathbf{z}^{T-1}\|^2 \leq \mathcal{O}\left(\frac{1}{T}\right)$$

Application 1: Constrained ℓ_1 problem

$$\boxed{\begin{aligned} \min \quad & \|\boldsymbol{x}\|_1 \\ \text{s.t.} \quad & \|\mathbf{A}\boldsymbol{x} - \mathbf{b}\|^2 \leq \delta \end{aligned}}$$

- Similar as the LASSO, but with explicit constrains on the noise magnitude
- Formulate into the ADMM form?
- Introduce new variables $\mathbf{z}_1, \mathbf{z}_2$

$$\begin{aligned} \min \quad & \|\boldsymbol{x}\|_1 \\ \text{s.t.} \quad & \|\mathbf{z}_1\|^2 \leq \delta \\ & \mathbf{z}_1 = \mathbf{A}\mathbf{z}_2 - \mathbf{b}, \ \mathbf{z}_2 = \boldsymbol{x} \end{aligned}$$

ADMM for Constrained ℓ_1 problem

- Formulate the augmented Lagrangian

$$L(\boldsymbol{x}, \mathbf{z}; \mathbf{y}) = \|\boldsymbol{x}\|_1 + \langle \mathbf{y}_1, \mathbf{z}_1 - \mathbf{A}\mathbf{z}_2 + \mathbf{b}\rangle + \frac{\rho}{2}\|\mathbf{z}_1 - \mathbf{A}\mathbf{z}_2 + \mathbf{b}\|^2$$
$$+ \langle \mathbf{y}_2, \mathbf{z}_2 - \boldsymbol{x}\rangle + \frac{\rho}{2}\|\mathbf{z}_2 - \boldsymbol{x}\|^2$$

- Put $(\boldsymbol{x}, \mathbf{z}_1)$ into one group (they are not "coupled")
- The $(\boldsymbol{x}, \mathbf{z}_1)$-step (proximity step + projection onto a ball)

$$\boldsymbol{x}^{r+1} = \arg\min_{\boldsymbol{x}} \|\boldsymbol{x}\|_1 + \langle \mathbf{y}_2^r, \mathbf{z}_2^r - \boldsymbol{x}\rangle + \frac{\rho}{2}\|\mathbf{z}_2^r - \boldsymbol{x}\|^2$$
$$\mathbf{z}_1^{r+1} = \arg\min_{\|\mathbf{z}_1\|^2 \le \delta} \langle \mathbf{y}_1^r, \mathbf{z}_1 - \mathbf{A}\mathbf{z}_2^r + \mathbf{b}\rangle + \frac{\rho}{2}\|\mathbf{z}_1 - \mathbf{A}\mathbf{z}_2^r + \mathbf{b}\|^2$$

- The $\mathbf{z}_2$-step (can be solved in closed form)

$$\mathbf{z}_2^{t+1} = \arg\min_{\mathbf{z}_2} \langle \mathbf{y}_1^r, -\mathbf{A}\mathbf{z}_2\rangle + \frac{\rho}{2}\|\mathbf{z}_1^{r+1} - \mathbf{A}\mathbf{z}_2 - \mathbf{b}\|^2 + \langle \mathbf{y}_2^r, \mathbf{z}_2\rangle + \frac{\rho}{2}\|\mathbf{z}_2 - \boldsymbol{x}^{r+1}\|^2$$

- Then update $\mathbf{y}_1, \mathbf{y}_2$ (easy)

ADMM for Constrained ℓ_1 problem - Summary

$$\boxed{\begin{aligned} \min \quad & \|\boldsymbol{x}\|_1 \\ \text{s.t.} \quad & \|\mathbf{A}\boldsymbol{x} - \mathbf{b}\|^2 \le \delta \end{aligned}}$$

- **Main difficulty**: objective nonsmooth + difficult constraint
- First step, introduce new variables to split these two things
- Apply ADMM, each "difficulty" is dealt with separably
- **Divide and Conquer**!

Application 2: Multiple Regularizations

- Consider the following sparse group LASSO problem
- The problem is formulated as

$$\min \frac{1}{2}\|\mathbf{A}\boldsymbol{x} - \mathbf{b}\|^2 + \underbrace{\sum_{g=1}^{G} \lambda_g \|\boldsymbol{x}_g\|_2}_{\ell_1/\ell_2 \text{norm}} + \underbrace{\lambda\|\boldsymbol{x}\|_1}_{\ell_1 \text{norm}}$$

- Select a few groups, and select a few variables within groups
- How to deal with two different norms simultaneously?

Application 2: Multiple Regularizations

- Same trick, do the variable splitting
- Introduce new variable $\mathbf{z}_1$, $\mathbf{z}_2$
- Perform the following reformulation

$$\begin{aligned} \min \quad & \frac{1}{2}\|\mathbf{A}\mathbf{z}_2 - \mathbf{b}\|^2 + \sum_{g=1}^{G} \lambda_g \|\boldsymbol{x}_g\|_2 + \lambda \|\mathbf{z}_1\|_1 \\ \text{subject to} \quad & \mathbf{z}_1 = \boldsymbol{x}, \quad \mathbf{z}_2 = \boldsymbol{x} \end{aligned} \tag{0.1}$$

- Reformulated in an ADMM form; deal with two norms separately?

ADMM for Multiple Regularizations

- Again let's first formulate the augmented Lagrangian

$$L(\boldsymbol{x}, \mathbf{z}; \mathbf{y}) = \frac{1}{2}\|\mathbf{A}\mathbf{z}_1 - \mathbf{b}\|^2 + \sum_{g=1}^{G} \lambda_g \|\boldsymbol{x}_g\|_2 + \lambda \|\mathbf{z}_2\|_1 + \langle \mathbf{y}_1, \mathbf{z}_1 - \boldsymbol{x} \rangle + \frac{\rho}{2}\|\mathbf{z}_1 - \boldsymbol{x}\|^2 + \langle \mathbf{y}_2, \mathbf{z}_2 - \boldsymbol{x} \rangle + \frac{\rho}{2}\|\mathbf{z}_2 - \boldsymbol{x}\|^2$$

- $\boldsymbol{x}$ is one variable bock, $(\mathbf{z}_1, \mathbf{z}_2)$ is a variable block
- $\boldsymbol{x}$ subproblem (proximal operator for group-Lasso)

$$\boldsymbol{x}^{r+1} = \arg\min \sum_{g=1}^{G} \lambda_g \|\boldsymbol{x}_g\|_2 + \frac{\rho}{2}\left\|\boldsymbol{x} - \mathbf{z}_1^r - \frac{1}{\rho}\mathbf{y}_1^r\right\|^2 + \frac{\rho}{2}\left\|\boldsymbol{x} - \mathbf{z}_2^r - \frac{1}{\rho}\mathbf{y}_2^r\right\|^2$$

- $\mathbf{z}_1$ subproblem easy (quadratic, unconstrained)
- $\mathbf{z}_2$ subproblem is proximal operator for Lasso

Application 3: Intersection of Two Sets

- Suppose we would like to find a vector that

$$\begin{aligned} & FIND \\ \text{subject to} \quad & \|\boldsymbol{x}\|_1 \le \delta, \quad \|\mathbf{A}\boldsymbol{x} - \mathbf{b}\|^2 \le \epsilon \end{aligned} \tag{0.2}$$

- Finding a intersection of two convex sets
- We can introduce two variables $\mathbf{z}_1$, $\mathbf{z}_2$

$$\begin{aligned} & FIND \\ \text{subject to} \quad & \|\mathbf{z}_1\|_1 \le \delta, \quad \|\mathbf{z}_2\|^2 \le \epsilon \\ & \mathbf{z}_1 = \boldsymbol{x}, \quad \mathbf{z}_2 = \mathbf{A}\boldsymbol{x} - \mathbf{b} \end{aligned}$$

- First block $\boldsymbol{x}$, second block $(\mathbf{z}_1, \mathbf{z}_2)$
- Dealing with each constraint separately

Application 3: Intersection of Two Sets

- The augmented Lagrangian

$$L(\boldsymbol{x}, \mathbf{z}; \mathbf{y}) = \langle \mathbf{z}_1 - \boldsymbol{x}, \mathbf{y}_1 \rangle + \frac{\rho}{2}\|\mathbf{z}_1 - \boldsymbol{x}\|^2 \\ + \langle \mathbf{z}_2 - \mathbf{A}\boldsymbol{x} + \mathbf{b}, \mathbf{y}_2 \rangle + \frac{\rho}{2}\|\mathbf{z}_2 - \mathbf{A}\boldsymbol{x} + \mathbf{b}\|^2$$

- $\mathbf{z}_1$ subproblem (projection on ℓ_1 ball, easy, see Here)

$$\min \quad \langle \mathbf{z}_1, \mathbf{y}_1^r \rangle + \frac{\rho}{2}\|\mathbf{z}_1 - \boldsymbol{x}^r\|^2, \quad \text{subject to } \|\mathbf{z}_1\|_1 \leq \delta$$

- $\mathbf{z}_2$ subproblem (projection on ℓ_2 ball, easy)

$$\min \quad \langle \mathbf{z}_2, \mathbf{y}_2^r \rangle + \frac{\rho}{2}\|\mathbf{z}_2 - \mathbf{A}\boldsymbol{x}^r + \mathbf{b}\|^2, \quad \text{subject to } \|\mathbf{z}_2\|_2^2 \leq \epsilon$$

- $\boldsymbol{x}$ subproblem, simple unconstrained quadratic problem, easy

Application 4: Consensus Problem

- Consider the following problem

$$\min \quad \sum_{i=1}^{N} f_i(\boldsymbol{x}) + R(\boldsymbol{x}), \text{ subject } \text{ to } x \in X \qquad (0.3)$$

- We can do the same procedure as before

$$\begin{aligned} \min \quad & \sum_{i=1}^{N} f_i(\mathbf{z}_i) + R(\boldsymbol{x}) \\ & \text{subject } \text{ to } \boldsymbol{x} \in X, \ \mathbf{z}_i = \boldsymbol{x}, \ \forall \ i \end{aligned}$$

- Each agent handles a single function f_i

ADMM for Consensus Problem

- Write down the augmented Lagrangian

$$L(\boldsymbol{x}, \mathbf{z}; \mathbf{y}) = \sum_{i=1}^{N} f_i(\mathbf{z}_i) + R(\boldsymbol{x}) + \sum_{i=1}^{N} \langle \mathbf{y}_i, \mathbf{z}_i - \boldsymbol{x} \rangle + \frac{\rho}{2} \|\mathbf{z}_i - \boldsymbol{x}\|^2$$

- $\boldsymbol{x}$ subproblem (proximity operator, at the control center)

$$\boldsymbol{x}^{r+1} = \arg\min \ R(\boldsymbol{x}) + \sum_{i=1}^{N} \langle \mathbf{y}_i^r, -\boldsymbol{x} \rangle + \frac{\rho}{2} \|\mathbf{z}_i^r - \boldsymbol{x}\|^2$$

- For each agent, $\mathbf{z}_i$ problem (at the agent)

$$\mathbf{z}_i^{r+1} = \arg\min \ f_i(\mathbf{z}_i) + \langle \mathbf{y}_i^r, \mathbf{z}_i \rangle + \frac{\rho}{2} \|\mathbf{z}_i - \boldsymbol{x}^{r+1}\|^2$$

- Update the dual variables (at the control center)

$$\mathbf{y}_i^{r+1} = \mathbf{y}_i^r + \rho(\mathbf{z}_i^{r+1} - \boldsymbol{x}^{r+1}), \ \forall \ i$$

ADMM for Consensus Problem (cont.)

- **Special Case**: Without the regularization term, we have

$$L(\boldsymbol{x}, \mathbf{z}; \mathbf{y}) = \sum_{i=1}^{N} f_i(\mathbf{z}_i) + \sum_{i=1}^{N} \langle \mathbf{y}_i, \mathbf{z}_i - \boldsymbol{x} \rangle + \frac{\rho}{2} \|\mathbf{z}_i - \boldsymbol{x}\|^2$$

- For each agent, $\mathbf{z}_i$ problem (at the agent)

$$\mathbf{z}_i^{r+1} = \arg\min \ f_i(\mathbf{z}_i) + \langle \mathbf{y}_i^r, \mathbf{z}_i \rangle + \frac{\rho}{2} \|\mathbf{z}_i - \boldsymbol{x}^r\|^2$$

- $\boldsymbol{x}$ subproblem (at the control center)

$$\boldsymbol{x}^{r+1} = \arg\min \sum_{i=1}^{N} \langle \mathbf{y}_i^r, -\boldsymbol{x} \rangle + \frac{\rho}{2} \|\mathbf{z}_i^{r+1} - \boldsymbol{x}\|^2 = \frac{1}{N} \sum_{i=1}^{N} (\mathbf{z}_i^{r+1} + \mathbf{y}_i^r/\rho)$$

- Update the dual variables (at the control center)

$$\mathbf{y}_i^{r+1} = \mathbf{y}_i^r + \rho(\mathbf{z}_i^{r+1} - \boldsymbol{x}^{r+1}), \ \forall \ i$$

- One can show (make sense ?)

$$\sum_{i=1}^{N} \mathbf{y}_i^{r+1} = 0 \longrightarrow \boldsymbol{x}^{r+1} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{z}_i^{r+1}$$

ADMM for Consensus Problem (cont.)

- Let $\bar{\mathbf{z}}^{r+1} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{z}_i^{r+1}$
- ADMM updates reduces to:

$$\mathbf{z}_i^{r+1} = \arg\min \; f_i(\mathbf{z}_i) + \langle \mathbf{y}_i^r, \mathbf{z}_i \rangle + \frac{\rho}{2}\|\mathbf{z}_i - \bar{\mathbf{z}}^r\|^2 \;\forall\; i$$

$$\mathbf{y}_i^{r+1} = \mathbf{y}_i^r + \rho(\mathbf{z}_i^{r+1} - \bar{\mathbf{z}}^{r+1}), \;\forall\; i$$

- Parallel execution by "gather-scatter"
  1. Gather $\mathbf{z}_i^r$ and average to get $\bar{\mathbf{z}}^r$
  2. Scatter the average $\bar{\mathbf{z}}^r$ to processors
  3. Compute $\mathbf{z}_i^{r+1}$ locally (in parallel)
  4. Compute $\mathbf{y}_i^{r+1}$ locally (in parallel)

Example: Consensus LASSO

- Lasso with consensus optimization (divide data by training samples) $\min_{\boldsymbol{x}} \frac{1}{2}\sum_{i=1}^N \|\mathbf{a}_i^T\boldsymbol{x} - b_i\|^2 + \lambda\|\boldsymbol{x}\|_1 = \min_{\mathbf{z}_i=\boldsymbol{x}} \frac{1}{2}\sum_{i=1}^N \|\mathbf{a}_i^T\mathbf{z}_i - b_i\|^2 + \lambda\|\boldsymbol{x}\|_1$
- Augmented Lagrangian

$$L = \sum_{i=1}^N \frac{1}{2}\|\mathbf{a}_i^T\mathbf{z}_i - b_i\|^2 + \lambda\|\boldsymbol{x}\|_1 + \langle \mathbf{y}_i, \mathbf{z}_i - \boldsymbol{x}\rangle + \frac{\rho}{2}\|\mathbf{z}_i - \boldsymbol{x}\|^2$$

- ADMM updates

$$\mathbf{z}_i^{r+1} = \frac{1}{2}\|\mathbf{a}_i^T\mathbf{z}_i - b_i\|^2 + \langle \mathbf{y}_i^r, \mathbf{z}_i\rangle + \frac{\rho}{2}\|\mathbf{z}_i - \boldsymbol{x}^r\|^2, \ \forall\ i$$

$$\boldsymbol{x}^{r+1} = \lambda\|\boldsymbol{x}\|_1 + \sum_{i=1}^N \langle \mathbf{y}_i^r, -\boldsymbol{x}\rangle + \frac{\rho}{2}\|\mathbf{z}_i^{r+1} - \boldsymbol{x}\|^2,$$

$$\mathbf{y}_i^{r+1} = \mathbf{y}_i^r + \rho(\mathbf{z}_i^{r+1} - \boldsymbol{x}^{r+1}), \ \forall\ i$$

Example: Consensus LASSO (from Boyd)

- Example with $\mathbf{A} \in \mathbb{R}^{400,000 \times 8000}$, roughly $30GB$ of data
- Split into 80 groups across 10 (each 8-core) machines
- Written in C using MPI, on Amazon EC2
- ADMM running time: 0.5-2 seconds
- Total running time, including loading data, etc, 5-6 mins.

Consensus SVM (from Boyd)

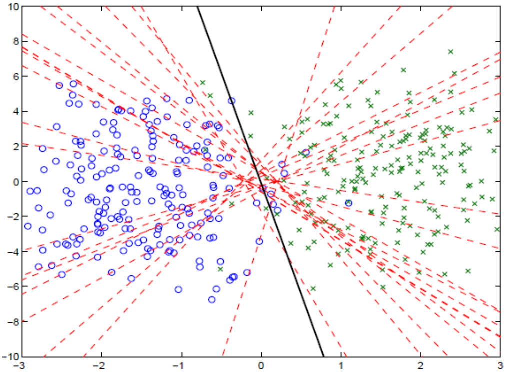
- Data $(\mathbf{a}_i, b_i), i = 1, \cdots, N$, $\mathbf{a}_i \in \mathbb{R}^K$, $y_i \in \{-1, +1\}$
- Linear SVM

$$\min_{\mathbf{w}} \frac{1}{N} \sum_{i=1}^{N} \max\left\{0, 1 - b_i \boldsymbol{x}^T \mathbf{a}_i\right\} + \frac{\lambda}{2} \|\boldsymbol{x}\|_2^2 \tag{0.4}$$

- Example: $K = 2$, $N = 400$, splits into 20 groups

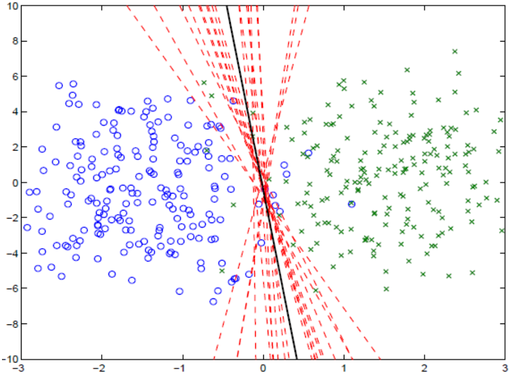
Consensus SVM (from Boyd)

Iteration 1



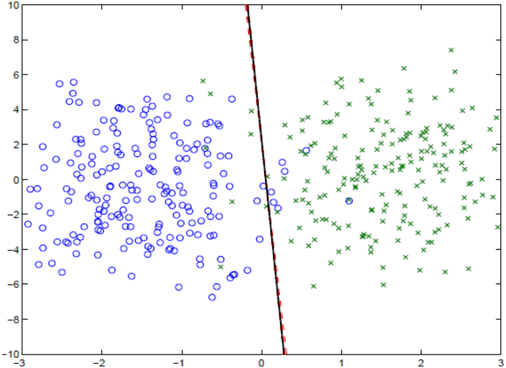
Consensus SVM (from Boyd)

Iteration 5



Consensus SVM (from Boyd)

Iteration 40



Outline

Problem Setup

$$\begin{array}{ll} \min & \|\mathbf{L}\|_* + \tau \|\mathbf{S}\|_1 \\ \text{subject to} & \|(\mathbf{L} + \mathbf{S} - \mathbf{M})(\Omega)\|_F^2 \leq \mu \end{array}$$

- Consider the video F-B separation problem
- Minimize low rank plus the sparse part
- Ω is the set of observed pixels (say 80% of the original pixels)
- $\mathbf{X}(\Omega)$ means the submatrix of $\mathbf{X}$ defined by the index set Ω
- The constraint means the difference between the reconstructed frames and the original frame on the observed pixels is small
- τ and μ are two known numbers (parameters of the problem)

ADMM Formulation

$$
\boxed{\begin{array}{rl} \min & \|\mathbf{L}\|_* + \tau\|\mathbf{S}\|_1 \\ \text{subject to} & \|(\mathbf{L}+\mathbf{S}-\mathbf{M})(\Omega)\|_F^2 \le \mu \end{array}}
$$

- Introduce a new variable $\mathbf{Z}$, consider

$$
\begin{array}{rl} \min & \|\mathbf{L}\|_* + \tau\|\mathbf{S}\|_1 \\ \text{subject to} & \|-\mathbf{Z}(\Omega)\|_F^2 \le \mu \\ & \mathbf{L}+\mathbf{S}+\mathbf{Z} = \mathbf{M} \end{array}
$$

- The augmented Lagrangian

$$
\begin{aligned} L(\mathbf{L},\mathbf{S},\mathbf{Z};\mathbf{\Lambda}) = \|\mathbf{L}\|_* + \tau\|\mathbf{S}\|_1 + \langle \mathbf{\Lambda}, \mathbf{L}+\mathbf{S}+\mathbf{Z}-\mathbf{M}\rangle \\ + \frac{\beta}{2}\|\mathbf{L}+\mathbf{S}+\mathbf{Z}-\mathbf{M}\|^2 \end{aligned}
$$

The ADMM: $\mathbf{Z}$-subproblem

- The augmented Lagrangian

$$L(\mathbf{L}, \mathbf{S}, \mathbf{Z}; \mathbf{\Lambda}) = \|\mathbf{L}\|_* + \tau\|\mathbf{S}\|_1 + \langle \mathbf{\Lambda}, \mathbf{L} + \mathbf{S} + \mathbf{Z} - \mathbf{M} \rangle + \frac{\beta}{2}\|\mathbf{L} + \mathbf{S} + \mathbf{Z} - \mathbf{M}\|^2$$

- Consider the $\mathbf{Z}$ subproblem
- Define $\mathbf{T} = \mathbf{M} + \mathbf{\Lambda}/\beta - \mathbf{L} - \mathbf{S}$
- The $\mathbf{Z}$-subproblem becomes

$$\min_{\mathbf{Z}} \|\mathbf{Z} - \mathbf{T}\|^2, \quad \text{subject to } \|\mathbf{Z}(\Omega)\|_F^2 \leq \mu$$

- The solution (again a projection operator)

$$\mathbf{Z}(\Omega) = \mathbf{T}(\Omega) \times \min(\sqrt{\mu}, \|\mathbf{T}(\Omega)\|_F)/\|\mathbf{T}(\Omega)\|_F$$

$$\mathbf{Z}(\bar{\Omega}) = \mathbf{T}(\bar{\Omega}) \quad \text{(don't care about the rest pixels)}$$

- Here $\mathbf{Z}(\Omega)$ means the components of $\mathbf{Z}$ corresponds to the set Ω

The ADMM: $\mathbf{S}$-subproblem

- The augmented Lagrangian

$$L(\mathbf{L}, \mathbf{S}, \mathbf{Z}; \mathbf{\Lambda}) = \|\mathbf{L}\|_* + \tau\|\mathbf{S}\|_1 + \langle \mathbf{\Lambda}, \mathbf{L} + \mathbf{S} + \mathbf{Z} - \mathbf{M}\rangle + \frac{\beta}{2}\|\mathbf{L} + \mathbf{S} + \mathbf{Z} - \mathbf{M}\|^2$$

- Consider the $\mathbf{S}$ subproblem
- Define $\mathbf{T} = \mathbf{M} + \mathbf{\Lambda}/\beta - \mathbf{L} - \mathbf{Z}$
- The $\mathbf{S}$-subproblem becomes

$$\min_{\mathbf{S}} \ \frac{\beta}{2}\|\mathbf{S} - \mathbf{T}\|^2 + \tau\|\mathbf{S}\|_1$$

- **Solution**: Soft-thresholding $\mathbf{S} = \mathrm{Soft}\left(\mathbf{T}, \frac{\tau}{\beta}\right)$

The ADMM: $\mathbf{L}$-subproblem

- The augmented Lagrangian

$$L(\mathbf{L},\mathbf{S},\mathbf{Z};\boldsymbol{\Lambda}) = \|\mathbf{L}\|_* + \tau\|\mathbf{S}\|_1 + \langle \boldsymbol{\Lambda}, \mathbf{L}+\mathbf{S}+\mathbf{Z}-\mathbf{M}\rangle + \frac{\beta}{2}\|\mathbf{L}+\mathbf{S}+\mathbf{Z}-\mathbf{M}\|^2$$

- Consider the $\mathbf{L}$ subproblem
- Define $\mathbf{T} = \mathbf{M} + \boldsymbol{\Lambda}/\beta - \mathbf{S} - \mathbf{Z}$
- The $\mathbf{L}$-subproblem becomes

$$\min_{\mathbf{S}} \; \frac{1}{2}\|\mathbf{L}-\mathbf{T}\|^2 + \frac{1}{\beta}\|\mathbf{L}\|_*$$

- **Solution**: Soft-thresholding on the singular values of $\mathbf{T}$

The ADMM: The dual Step

- The final step is to update the dual variable

$$\mathbf{\Lambda} = \mathbf{\Lambda} + \beta(\mathbf{L} + \mathbf{S} + \mathbf{Z} - \mathbf{M})$$

- When to stop? Two criteria
  1. The following quantity is small enough

$$\max\left\{\|\mathbf{L}^{r+1} - \mathbf{L}^{r}\|/(1 + \|\mathbf{L}^{r+1}\|), \|\mathbf{S}^{r+1} - \mathbf{S}^{r}\|/(1 + \|\mathbf{S}^{r+1}\|)\right\}$$

  2. The following quantity is small enough

$$|f(\mathbf{L}^{r+1}, \mathbf{S}^{r+1}) - f(\mathbf{L}^{r}, \mathbf{S}^{r})|/|f(\mathbf{L}^{r+1}, \mathbf{S}^{r+1})|$$

  3. Performance see e.g. `https://arxiv.org/pdf/1712.06229`